Exp: 1

Network Commands used in Linux and windows

Aim:

To study and execute basic networking commands in windows and Linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status and troubleshooting network-related issue

Windows Commands:

i) ipconfig

Displays

* IP address
* Subnet mask
* Default Gateway

Sample Output

IPv4 Address ... : 192.168.1.20
Subnet Mask .... : 255.255.255.0
Default Gateway ... : 192.168.1.1)

ii) ipconfig /all

Shows

* MAC Address
* DHCP Server
* DNS server

```
[sj@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1f:0d:cc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86069sec preferred_lft 86069sec
    inet6 fe80::a00:27ff:fe1f:dcc/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[sj@localhost ~]$
```

```
[sj@localhost ~]$ hostname
localhost.localdomain
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ping google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=118 time=6.27 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=118 time=7.93 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=118 time=11.1 ms
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=118 time=5.11 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=118 time=12.3 ms
```

```
[sj@localhost ~]$ nslookup google.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.110
Name:   google.com
Address: 2404:6800:4007:815::200e

[sj@localhost ~]$
```

```
[sj@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
```

```
[sj@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:1f:0d:cc brd ff:ff:ff:ff:ff:ff
[sj@localhost ~]$
```

* Lase information
* Adapter Details

iii) ping google.com

(or)

ping 8.8.8.8

Purpose

Checks whether the destination is reachable

Sample output:

Reply from: 8.8.8.8

Packets: sent = 4

Recivad = 4

Lost = 0

iv) tracert google.com

purpose

Displays every router (hop) between your Computer and destination

Sample output

1. 192.168.1.1

2 ISP Router

3 Google Network

v) nslookup google.com

purpose

```
[s]localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:2f:d8:cc brd ff:ff:ff:ff:ff:ff
[s]localhost ~]$
[s]localhost ~]$ ss -tuln
Netid  State      Recv-Q Send-Q                Local Address:Port                 Peer Address:Port
udp    UNCONN     0      0                     10.0.2.15%enp0s3:68                10.0.2.2:67
[s]localhost ~]$
```

```
[s]localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

Displays

* DNS Server
* IP Address of website

Sample output

Name : google.com

Address : 142. 250. xxx. xxx

vi) route print

purpose

Displays

* Network Routes
* Gateway
* Destination Network

vii) netstat

purpose

Displays

* Active TCP Connections
* Listening ports

viii) netstat -a

Shows

* All ports
* TCP connections
* UDP ports

ix) getmac

purpose

Displays MAC address of the computer

x) hostname

purpose

Displays Computer name

Eg:

DESKTOP-ABC123

xi) arp -a

purpose

Displays IP-to-MAC address mappings

Result:

Basic networking commands were successfully executed on Linux and Windows IP addresses, routing tables, ARP caches, and DNS resolution were observed and recorded